Resilient I-Bug Navigation under Intermittent Signal Spoofing

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Abstract

Minimalist autonomous robots operating in GPS-denied environments rely on local sensor feedback rather than global coordinates or odometry. The I-Bug algorithm achieves provable target convergence using only a contact sensor and a scalar signal-intensity sensor. However, in noncoherent sensing settings where the receiver observes additive contributions from multiple emitters, intensity-based navigation is acutely vulnerable to intermittent adversarial spoofing. An additive spoofing transmitter warps the spatial intensity gradient in free space and injects false maxima along obstacle boundaries, trapping standard I-Bug agents in perpetual loops. We present a kinematics-based temporal anomaly filter and two counter-strategies—a *Cautious* strategy that conditionally preserves the convergence guarantee of benign I-Bug under our threat model, and a low-latency *Inertial* heuristic—requiring zero additional hardware. We prove a conditional preservation theorem showing that under intermittent step-spoofing (Assumptions 1–2), the Cautious strategy preserves the exact active-state spatial trajectory of any converging benign I-Bug execution, and we prove that freezing memory alone while continuing deterministic boundary motion is insufficient for universal convergence due to topological phase-locking (resonance). Paired evaluations across $1,000,000$ randomized 2D obstacle environments show that under the sampled environment distribution, the unmodified Naive I-Bug baseline succeeds in only $54.68\% \pm 0.10\%$ of runs, the Inertial heuristic achieves $99.61\% \pm 0.01\%$ success while reducing mean steps among successful runs by 20.5%, and the Cautious strategy achieves a 100.00% observed success rate (95% Clopper–Pearson CI: $[99.9996\%, 100.00\%]$), matching the benign (no-attack) I-Bug reference on the same mazes at a mean cost of 60.4 additional steps.

1 Introduction and Threat Model

Classical Bug navigation provides convergence guarantees in unknown planar environments using limited local sensing [1, 2], while the I-Bug algorithm replaces coordinate and range information with target-signal intensity [3, 4]. At this theoretical limit of sensory minimalism, a disk-shaped robot of radius $r_c > 0$ navigates a bounded planar workspace $\mathcal{W} \subset \mathbb{R}^2$ containing unknown, piecewise-analytic obstacles $\mathcal{O} = \{O_1, \ldots, O_M\}$ (with collision-free configuration space $\mathcal{F} \subset \mathcal{W}$) using only a local contact bumper and a scalar sensor measuring emission intensity from an unknown target beacon $p_t \in \mathcal{F}$.

In benign conditions, the target intensity is a strictly decreasing radial function of Euclidean distance, $m(p - p_t) = \|p - p_t\|_2^{-1}$, and the mission succeeds when $\|p - p_t\|_2 < R_{\text{reach}}$. Without coordinates or odometry, the robot executes three primitives at discrete control steps $k \in \mathbb{N}_0$ (step period Δt, step size $\Delta s = v_{\max}\Delta t$):

1. u_{ori} (**Orient**): Aligns heading θ_k with a local gradient-direction estimate obtained from spatially local queries of the same scalar intensity sensor $S(p,t)$:
$$\theta_\nabla(p,t) = \operatorname{atan2}\big(\partial_y S(p,t),\, \partial_x S(p,t)\big) \pmod{2\pi}. \tag{1}$$
This realizes the I-Bug orientation primitive without an additional sensor: in simulation, the derivatives are central finite differences of S at four nearby probe points within one control step (we assume $u(t)$ does not switch during the probes).

2. u_{fwd} (**Move Forward**): Steps linearly along θ_k while in free-space state `MoveToTower`. Upon obstacle contact, it records threshold $i_H \leftarrow \max(S(p_k, t_k), i_H)$, turns right by $\pi/2$, and enters `FollowObstacle`.

3. u_{fol} (**Follow Boundary**): Tracks the obstacle perimeter keeping the wall on its left while updating the local boundary maximum $i_{\max}$. Once $i_{\max} > i_H + \epsilon$ followed by a descending sample $S(p_k, t_k) < i_{\max} - \epsilon$, it updates $i_H \leftarrow i_{\max}$, applies u_{ori}, and returns to `MoveToTower`.

Adversarial Additive RSS Model. Spoofed navigation signals can manipulate autonomous platforms, while sophisticated attackers may employ intermittent or covert signal takeover to disrupt guidance [5, 6]. Consider a malicious transmitter at an unknown location $p_e \in \mathcal{F}$ broadcasting an intermittent spoofing signal of strength $\alpha_e > 0$ governed by switching indicator $u(t) \in \{0, 1\}$.

Assumption 1 (Additive RSS Superposition Field)**.** *We study an incoherent additive received-signal-strength (RSS) abstraction in which the sensor reports the sum of the legitimate and adversarial source contributions at position $p \in \mathcal{F}$ and time $t \geq 0$:*
$$S(p,t) = m(p - p_t) + e(p,t), \quad e(p,t) = u(t)\,\frac{\alpha_e}{\|p - p_e\|_2}. \tag{2}$$

When $u(t) = 1$, $e(p,t)$ corrupts both the scalar reading $S(p,t)$ and its spatial gradient:
$$\nabla_p S(p,t) = -\frac{p - p_t}{\|p - p_t\|_2^3} - u(t)\,\alpha_e \frac{p - p_e}{\|p - p_e\|_2^3}. \tag{3}$$

Thus, an unprotected robot is deflected off course during u_{ori} and records poisoned hit thresholds i_H during obstacle contact that cannot be satisfied once $u(t) \to 0$, trapping the robot in infinite boundary-following loops.

2 Physics-Informed Anomaly Detection and Counter-Strategies

Motivated by physics-based attack detection [7] and classical abrupt-change detection [8], we defend the robot without additional hardware or oracle access to $m(p-p_t)$ by exploiting the separation between bounded motion-induced variation and discontinuous transmitter switching.

Assumption 2 (Kinematic Lipschitz Bound vs. Temporal Discontinuity)**.** *Over a high-rate sensor sampling interval $\delta t \ll \Delta t$ (maximum displacement $\delta s = v_{\max}\delta t \ll R_{reach}$) during which the adversary is inactive ($u \equiv 0$), the natural intensity variation from reference $I_{ref} = m(p-p_t) = 1/d$ is bounded by the local radial Lipschitz increment:*

$$\Delta I_{\max}(I_{ref}) = \frac{\delta s}{d(d-\delta s)} = \frac{\delta s\, I_{ref}^2}{1-\delta s\, I_{ref}}. \tag{4}$$

Conversely, $u(t)$ switches between 0 and 1 such that every active burst lasts at most $T_{on} \geq 1$ steps and every inactive interval lasts at least $T_{off} > N_{confirm}$ steps (a fixed-period schedule is a special case), producing an instantaneous step jump

$$e_{\min} := \frac{\alpha_e}{\max_{p\in\mathcal{F}_{nav}} \|p-p_e\|_2} \geq \frac{\alpha_e}{\operatorname{diam}(\mathcal{W})} \tag{5}$$

that satisfies $e_{\min} > \eta_{spike} \max_{p\in\mathcal{F}_{nav}} \Delta I_{\max}(m(p-p_t))$.

Remark 1 (Physical Motivation, Scope & Bounded Noise)**.** *Assumption 2 is motivated by time-scale separation: electronic sampling (kHz) is orders of magnitude faster than mechanical translation ($v_{\max}$), making $\delta s = v_{\max}\delta t$ and $\Delta I_{\max}$ small relative to switching transients. The guarantee excludes continuously active, slowly ramped, or sub-threshold attacks whose temporal residual remains below the detection threshold [6]; formal analysis of such stealthy or near-field attacks is left for future work. In particular, a defense-aware adversary can stall the Cautious robot indefinitely (denial of service) by keeping $u(t) = 1$ or switching off for at most $N_{confirm}$ steps; Theorem 1 guarantees safety, not liveness, against such attackers. Under bounded noise $|\nu_k| \leq \varepsilon_{noise}$, false positives are excluded if $\Delta I_{\max} + 2\varepsilon_{noise} < \eta_{spike}\Delta I_{\max}$, and false negatives are excluded if $e_{\min} - 2\varepsilon_{noise} > \eta_{spike}\Delta I_{\max}$ (and analogously for $\mathcal{A}_{drop}$ with $\eta_{drop} \leq \eta_{spike}$).*

Anomaly Filter. At the start of step k (before actuation), the filter compares the pre-step reading $I_k^- = S(p_{k-1}, t_k)$ against the previous post-step reading $I_{k-1}^+ = S(p_{k-1}, t_{k-1})$. Note that at an activation transition ($0 \to 1$), $I_{k-1}^+ = m(p_{k-1}-p_t)$ is uncorrupted, whereas at a deactivation transition ($1 \to 0$), $I_k^- = m(p_{k-1}-p_t)$ is uncorrupted. With $\Delta I_k = I_k^- - I_{k-1}^+$, positive (*spike*) and negative (*drop*) anomalies are flagged via:

$$\mathcal{A}_{\text{spike}} \iff \Delta I_k > \eta_{\text{spike}}\, \Delta I_{\max}(I_{k-1}^+), \tag{6}$$

$$\mathcal{A}_{\text{drop}} \iff (-\Delta I_k > 0.15 I_{k-1}^+) \vee (-\Delta I_k > \eta_{\text{drop}} \Delta I_{\max}(I_k^-)), \tag{7}$$

where $\eta_{\text{spike}} = 2.0$ and $\eta_{\text{drop}} = 1.2$. Both readings are taken at the same position p_{k-1}, so the benign ΔI_k is exactly zero in our noiseless simulation; $\Delta I_{\max}$ is a conservative margin for physical sensors that integrate over δt while moving up to δs, and for noise (Remark 1). The 15% term is a heuristic safeguard that only adds detections and is not used by the analysis. To the best of our knowledge, we are not aware of a prior anti-spoofing mechanism designed specifically for the minimalist I-Bug framework; we evaluate three strategies:

- **Naive (Original I-Bug Baseline [4]):** Bypasses the filter, executing $u_{\text{ori}}, u_{\text{fwd}}, u_{\text{fol}}$ and updating i_H directly on $S(p,t)$.
- **Cautious (State-Freezing):** Upon $\mathcal{A}_{\text{spike}}$, the robot enters `Paused` mode: it commands zero velocity ($\Delta p_k = 0$), locks its pose ($p_{\text{frozen}}, \theta_{\text{frozen}}$) and internal memory ($i_H, i_{\max}, i_{\text{mem}}$), and waits until $\mathcal{A}_{\text{drop}}$ is observed and confirmed over $N_{\text{confirm}} = 3$ consecutive stable steps before resuming `Normal` closed-loop navigation.
- **Inertial (Open-Loop Heuristic):** Upon $\mathcal{A}_{\text{spike}}$, the robot enters `InertialMode`: it freezes its heading θ_{frozen} and suspends memory/peak updates, but continues moving. In `MoveToTower`, it suppresses u_{ori} and steps forward linearly via u_{fwd} (initializing $i_H \leftarrow i_{\text{mem}}$ from its last clean memory if an obstacle is hit while blind). In `FollowObstacle`, it continues perimeter tracking via u_{fol} while suspending the leave-obstacle check until $\mathcal{A}_{\text{drop}}$ restores `Normal` mode.

3 Formal Convergence & Resonance Analysis

Theorem 1 (Conditional Active-State Trajectory and Convergence Preservation)**.** *Under Assumptions 1 and 2 with $\eta_{drop} \leq \eta_{spike}$ and $T_{off} > N_{confirm}$, for any workspace $\mathcal{W}$ and benign field m: (i) after removing the paused steps, the sequence of poses and internal states of the Cautious robot coincides step-by-step with the benign baseline I-Bug execution γ_{ideal} (paused steps only repeat the last active pose); and (ii) whenever the benign execution reaches $\|p_k - p_t\|_2 < R_{reach}$ in $N_{ideal} < \infty$ active steps (in particular, under the convergence assumptions of [4]), the Cautious robot reaches the target in at most*

$$N_{cautious} \leq N_{ideal} + \left\lceil \frac{N_{ideal}}{T_{off} - N_{confirm}} \right\rceil (T_{on} + N_{confirm}) \tag{8}$$

finite steps.

Proof. At every adversarial activation step k_j^{on}, $\Delta I_k = m_e(p_{k-1} - p_e) \geq e_{\min} > \eta_{\text{spike}} \Delta I_{\max}(I_{k-1}^+)$, triggering $\mathcal{A}_{\text{spike}}$ prior to actuation or memory modification. Throughout $[k_j^{\text{on}}, k_j^{\text{off}} - 1]$, p_k is stationary and $\Sigma_k = (p_k, \theta_k, \text{mode}, i_H, i_{\max}, i_{\text{mem}})$ remains locked. At k_j^{off}, since $\eta_{\text{drop}} \leq \eta_{\text{spike}}$, the drop $-m_e(p_{k-1} - p_e) \geq e_{\min} > \eta_{\text{drop}} \Delta I_{\max}(I_{k_j^{\text{off}}}^-)$ triggers $\mathcal{A}_{\text{drop}}$, and after N_{confirm} stationary verification ticks with $u(t) = 0$, the robot resumes at $k_j^{\text{res}} = k_j^{\text{off}} + N_{\text{confirm}} - 1$ with $\Sigma(k_j^{\text{res}}) = \Sigma(k_j^{\text{on}} - 1)$. Consequently, during every active step τ, $u(t_k) = 0 \implies S(p, t_k) = m(p - p_t)$ and $\nabla_p S = \nabla_p m$. By induction on active steps τ, $\Sigma(\tau) \equiv \Sigma_{\text{ideal}}(\tau)$, so the active-step subsequence of γ_{cautious} equals γ_{ideal}. Whenever γ_{ideal} converges in $N_{\text{ideal}} < \infty$ steps, each switching cycle yields at least $T_{\text{off}} - N_{\text{confirm}} \geq 1$ active steps for at most $T_{\text{on}} + N_{\text{confirm}}$ paused steps, establishing Eq. (8). □

Proposition 1 (Insufficiency of Memory Freezing Alone)**.** *Freezing internal memory alone while continuing deterministic boundary motion is insufficient to guarantee universal convergence. Let $\mathcal{A}_L \subset \partial O_i$ denote the boundary departure-detection arc over which the leave-event sequence (climb above $i_H + \epsilon$, peak $i_{\max}$, and descent by ϵ) must be evaluated on an obstacle O_i of perimeter L_i. If the orbital period $T_{orbit} = L_i/(v_{fol}\Delta t)$ satisfies $T_{orbit} = m\,T_{cycle}$ ($m \in \mathbb{N}$) and the blind arc of length $v_{fol}T_{on}\Delta t$ during $[k_j^{on}, k_j^{off})$ covers $\mathcal{A}_L$, the Inertial robot will bypass $\mathcal{A}_L$ while peak detection is suspended on every subsequent revolution, trapping it in an infinite limit cycle (while near-commensurate periods $T_{orbit} \approx m\,T_{cycle}$ induce prolonged finite trapping).*

4 Large-Scale Empirical Evaluation

We implemented a continuous 2D kinematic simulator in Julia ($\mathcal{W} = [0, 20]^2$, $p_0 = (2, 1)$, $p_t = (18, 18)$, $R_{\text{reach}} = 0.5$, $r_c = \Delta s = 0.3$, $T_{\text{cycle}} = 40$, $T_{\text{on}} = 10$, $\alpha_e = 1.0$, horizon $N_{\max} = 2{,}000$, filter $\delta s = 0.002$). The transmitter p_e is sampled uniformly in free space with $\|p_e - p_0\|_2 > 6$, $\|p_e - p_t\|_2 > 8$, and follows a fixed schedule ($T_{\text{on}} = 10$, $T_{\text{off}} = 30$, first activation at step 20). Across $1{,}000{,}000$ procedurally generated mazes (10–16 overlapping rectangular walls with wide collision-free passages and perimeter margins), all three strategies were evaluated in paired trials, plus a no-attack ($u \equiv 0$) benign I-Bug reference on the same mazes.

Table 1: Paired Evaluation (10^6 Runs; Step Stats on Succ. vs. Capped All)

Strategy	Success (95% CI)	Mean (Succ.)	Med. [IQR] (Succ.)	$\bar{N}_{\text{capped}}$
Benign I-Bug (no attack)	100.00%	132.6	126.0 [108, 150]	132.6
Naive [4]	54.68% ± 0.10%	199.3	153.0 [121, 205]	1015.4
Cautious	**100.00**% ([99.9996, 100])	193.0	178.0 [160, 215]	193.0
Inertial	99.61% ± 0.01%	**153.4**	**135.0** [115, 168]	**160.6**

Results and Conclusion. Table 1 and Fig. 1 corroborate our theoretical findings under the sampled environment distribution: (1) The unmodified *Naive* I-Bug baseline fails in 45.32% of the sampled environments, with representative failures exhibiting gradient deflection and threshold poisoning (e.g., $i_H = 1.230$ in Fig. 1). (2) Under Assumptions 1–2, the *Cautious* strategy achieves an observed 100.00% success rate ($10^6/10^6$ runs, maximum observed completion time of 578 steps; 137 steps in Fig. 1), corroborating Theorem 1. Since benign I-Bug also succeeds in every maze, this shows *preservation* rather than improvement: Cautious succeeds exactly where benign I-Bug does, satisfies Eq. (8) in every run, and pays a mean overhead of 60.4 steps (+45.6%). (3) Among successful runs, the *Inertial* heuristic reduces mean completion steps by 20.5% (153.4 vs. 193.0 steps; and by 16.8% under the unconditional horizon-capped metric $\bar{N}_{\text{capped}} = 160.6$), while failing in 0.39% ($3{,}877$) of trials; its observed failures are consistent with the phase-locking mechanism of Proposition 1, as illustrated in Fig. 1 ($i_H = 0.167$). This establishes a clear security-efficiency trade-off: full state-and-motion freezing conditionally preserves benign I-Bug convergence under intermittent step-spoofing (excluding continuously active, slowly ramped, or sub-threshold attacks), whereas open-loop continuation maximizes speed when rare resonance traps are tolerable or mitigated via stochastic excitation.

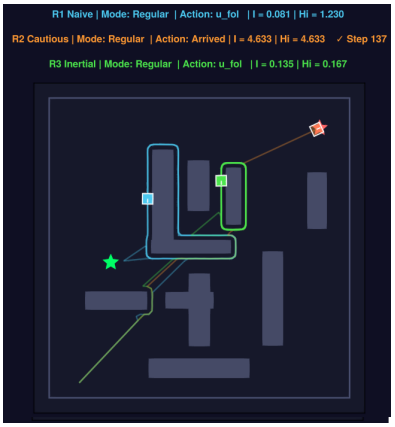


Figure 1: Representative double failure: Naive is threshold-poisoned at the L-obstacle; Inertial enters a phase-locked boundary loop; Cautious pauses during spoofing and reaches the target.

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